Airline Loyalty Intelligence

Identifying Silent Churn and Building a Segment-Specific Retention Engine

Prepared by: Krishna Vijay Kunwar (IIT Madras | Chemical Engineering, BIT Mesra)
Prepared for: Consulting & Analytics Club Evaluation Committee, IIT Guwahati

1. Executive Summary

Every loyalty program carries a hidden liability that never appears on a balance sheet: members who are technically “active” but have, in every practical sense, already left. Our analysis of 16,737 Canadian loyalty members confirmed this gap. The legacy Customer Lifetime Value (CLV) model describes who has been valuable, but it cannot say who is about to stop being valuable until the damage is already visible in next year's numbers.

We call this gap Silent Attrition. To close it, we built a behavioural-and-demographic segmentation, a calibrated machine learning early-warning model, and a segment-specific retention playbook. Three findings anchor this report:

- A clear, high-risk segment exists: At-Risk Infrequent Flyers (3,285 members, 26% of the active base) show a 10.6% Silent Churn rate - roughly 5× every other segment. The driver is not income or marital status, but inconsistent month-to-month activity.
- Our behavioural churn definition is validated against the dataset's own formal cancellation records: 92.4% of members we flag as Silent Churn go on to formally cancel - confirming this is a genuine early-warning signal, not an arbitrary proxy.
- The early-warning model is honest, not inflated: 0.77 out-of-sample ROC-AUC, with probability calibration explicitly checked and corrected (a 0.234 calibration gap was found and reduced to 0.0085) before any score was used for targeting.

We recommend three segment-specific interventions, detailed in Section 6, prioritized by a combination of churn risk and revenue exposure.

2. Framing the Problem: Silent Churn vs. Formal Cancellation

Airline loyalty is structurally different from subscription churn. There is no bill to stop paying and no contract to lapse - a member can simply stop flying, and their account remains “active” indefinitely. Formal cancellation, the metric most legacy systems track, captures only members who take a deliberate administrative step - and by then, the relationship was typically already cold for a long time.

We define Silent Churn as: a member who flew in 2017 and recorded zero flights in 2018. This is directly measurable from behaviour, with no need to wait for an administrative event.

Critically, we did not adopt this definition on faith - we tested it against the dataset's actual formal cancellation records:

	Not Formally Cancelled	Formally Cancelled
No Silent Churn	11,489	397
Silent Churn	42	511

n = 12,439 members active in 2017.

92.4% of members flagged as Silent Churn go on to formally cancel, confirming this is a real, predictive leading indicator. A smaller group (397 members) formally cancelled without showing Silent Churn in this specific window - likely cancelling in an earlier year or stopping abruptly outside our observation period. We proceed with Silent Churn as the model's target because it is observable before the lagging cancellation event, which is the entire purpose of an early-warning system, while stating this known limitation explicitly.

3. Data Preparation & Known Issues

Before modelling, we audited the raw data for the anomalies any real-world dataset contains:

- Negative salary values: 20 records showed a negative salary (minimum: − $58,486) - a data entry error, not a valid observation. These were treated as missing, not averaged in as-is.
- Missing salary: 4,258 records (including the corrected negative values) were missing salary entirely. Imputed with the median ($73,510), with an explicit flag column preserved for auditability - not silently filled and forgotten.
- Column-naming verification: the official Data Dictionary specifies the flight-volume field as “Total Flights.” We verified this explicitly against the real file before building any feature on top of it, since an unverified assumption here would silently break the entire pipeline.

With these corrections applied, our active analysis base is 12,439 members - those who flew at least once in 2017 and therefore represent the airline's addressable, analysable market for this study.

4. The Temporal Firewall: Preventing Data Leakage

The single most important structural decision in this project was enforcing what we call a Temporal Firewall. Every input features the model is allowed to see comes exclusively from 2017 - flight frequency, distance, points activity, tenure, salary, education, and segment. The model is trained to predict one single outcome: did this member go quiet in 2018?

Without this discipline, it is easy to accidentally let a model “see the future” - for example, by allowing 2018 flight counts to influence a feature used to predict 2018 churn. Such a model looks excellent in testing and fails completely in production, because in the real world you never have next year's data when making this year's decision. The Temporal Firewall guarantees our reported model performance is an honest preview of real deployment performance.

5. Behavioural + Demographic Segmentation

CLV is a useful number, but it only tells the CFO how much a member is worth - not why, or what would keep them engaged. Two members can carry similar CLV for entirely different reasons, and the same retention offer that delights one may fall flat for the other.

We applied K-Means clustering across both behavioural signals (2017 flights, distance, CLV) and demographic signals (salary, education level, marital status, loyalty card tier) together - not behavioural data alone. The number of clusters was selected using silhouette score analysis, then stability-tested across 5 independent random seeds before being finalized, rather than assumed in advance.

This process identified 5 statistically robust, demographically distinct segments:

Segment	Size	Avg CLV	Avg Salary	Churn Rate
At-Risk Infrequent Flyers	3,285	$6,445	$74,127	10.6%
High-Value Frequent Flyers	846	$26,912	$72,836	4.3%
Elite Professionals (Low-Freq.)	428	$7,968	$202,377	3.3%
Single Frequent Flyers	3,565	$6,400	$73,909	2.1%
Married Loyalists	4,315	$6,575	$73,888	1.8%

Source: K-Means clustering, k=5 (silhouette-validated, n=5 seed stability-tested), n=12,439.

Two findings stand out. First, At-Risk Infrequent Flyers show a churn rate roughly 5× higher than any other segment - and this is driven almost entirely by inconsistent monthly activity, not income or marital status (both of which sit close to the population average for this segment). Second, Elite Professionals is a small but genuinely distinct niche: 428 members, 100% holding a Doctorate or advanced degree, with nearly triple the average salary of every other segment - yet only moderate CLV, since their value comes from income and status rather than flight volume. A generic “fly more, earn more” incentive would likely miss this group entirely.

6. The Early Warning Model & Calibration

An XGBoost classifier was trained on the full 2017-only feature set (behavioural + demographic + segment), achieving a 0.7733 out-of-sample ROC-AUC. The single most important feature, by a wide margin, was active_months_2017 - how consistently a member flew throughout the year, not simply their total flight count. This is an intuitive and explainable result: a member who flies the same total distance in one trip versus spread across eight separate months has a very different relationship with the program.

Before using this model's output for any targeting decision, we checked whether its predicted probabilities could actually be trusted as real risk percentages - a step many churn models skip. We found a meaningful calibration gap: the model's predicted risk in its highest-risk decile was 79.8%, while the true observed churn rate in that decile was only 16.9%. Left uncorrected, this would have led the business to believe its highest-risk members were far more likely to churn than they actually were.

We corrected this using isotonic regression, fit on a genuine third data split held separate from both training and evaluation. After correction, the average calibration gap fell from 0.234 to 0.0085 - meaning a member now shown as “15% risk” in the dashboard genuinely churns at approximately 15% in validation, not an inflated or arbitrary score.

7. The Operational Playbook: Segment-Specific Interventions

A churn score alone tells us a member is at risk; the segment tells us which underlying problem is driving that risk, and therefore which response is likely to work. We recommend the following as Phase 1 interventions, prioritized by combined risk and revenue exposure:

Segment	Risk Driver	Recommended Action
At-Risk Infrequent Flyers	Inconsistent monthly activity - the strongest churn predictor in the model, well ahead of any demographic trait	Automated re-engagement nudge timed to each member's historical low-activity months, paired with a modest bonus-miles offer. Highest priority: largest segment, highest churn rate.
High-Value Frequent Flyers	Highest CLV segment; even a small dip in frequency is meaningful given revenue concentration	Proactive account manager check-in triggered by any early frequency dip, even before the model flags high risk - justified by this segment's outsized value.
Elite Professionals (Low-Freq.)	Engagement driven by income/status, not travel volume - a flight-based incentive may simply not be relevant to this group	Status-tier perks unrelated to flight volume (lounge access, partner benefits) rather than mileage-based offers.

Married Loyalists and Single Frequent Flyers show low churn (1.8% and 2.1%) and are recommended for standard-cadence retention only — intensive intervention budget is better spent on the three segments above.

8. Geographic & Enrolment Channel Analysis

We additionally tested whether geography (Province) or enrolment channel showed meaningful churn variation - in the interest of surfacing any pattern a summary report might miss, per the brief's evaluation standard, even where the result is a modest or null finding rather than a dramatic one.

Province	Members	Churn Rate	Avg CLV
Manitoba (highest)	489	6.3%	$8,007
Ontario (largest)	4,043	4.5%	$7,860
Yukon (lowest)	86	2.3%	$6,866

Full 11-province breakdown available in the companion notebook output.

Geographic variation exists (2.3%–6.3% range across provinces) but is modest compared to the behavioural segmentation finding (1.8%–10.6% across our 5 segments). We do not recommend province-specific intervention design on this basis alone - the signal is real but secondary to the behavioural driver identified in Section 5.

Enrolment channel showed no variation at all: every member in our active 2017 analysis base enrolled through the “Standard” channel. This is a clean, honest null result, not an oversight - it likely reflects that the dataset's “2018 Promotion” enrolment cohort had not yet accumulated 2017 flight activity by the nature of when they joined, and therefore falls outside our analysable base entirely. We flag this rather than omit the variable silently, since a reviewer checking enrolment-channel effects should know this was tested and why no result is shown.

9. Recommended Monitoring Metrics

Beyond the initial Phase 1 interventions, we recommend the following as standing metrics for the marketing and analytics team to track on an ongoing basis:

- Active-months trend by segment: since inconsistent monthly activity is the model's strongest predictor, a rolling view of this metric by segment provides the earliest possible warning - earlier than waiting for a full-year silent churn event.
- Model calibration drift: re-run the calibration check (Section 6) each time the model is retrained on a new year of data. A widening gap between predicted and actual churn rates is a sign the model needs refreshing, independent of its AUC.
- Segment migration: track what share of members moves from Married Loyalists or Single Frequent Flyers into the At-Risk Infrequent Flyers segment year over year - this would surface broader engagement erosion before it shows up in aggregate churn statistics.
- Playbook conversion rate: for each of the three Phase 1 interventions, track what share of contacted at-risk members shows renewed activity in the following quarter - this validates (or challenges) the recommended actions with real outcomes, not just model scores.

10. Limitations & Honest Caveats

- Silent Churn is a strong but imperfect proxy for true disengagement: 397 formally-cancelled members did not show Silent Churn in our 2017– 2018 window, meaning some cancellations originate outside this specific observation period.
- The model's 0.77 AUC reflects genuine irreducible uncertainty in human behaviour - we did not chase an artificially higher number by overfitting; this is a defensible, real ceiling given the feature set.
- Calibration was validated on a single held-out split; a production deployment would benefit from periodic recalibration as new data accumulates, particularly if member behaviour patterns shift over time.
- Demographic features (salary, education) are self-reported at enrolment and may not reflect a member's current circumstances.
- This analysis covers only 2017–2018 - a single year-over-year transition. The Temporal Firewall methodology is designed to be reusable as new years of data accumulate, but the specific patterns reported here (e.g., the exact churn rates by segment) should be re-validated once a second transition year is available; to confirm they are not specific to this one period.
- The clustering and model were both built on the same 12,439-member active base; while we used separate train/test/calibration splits for the model itself, the segmentation was not independently re-validated on a held-out sample. A future iteration could test segment stability on a subsequent year's cohort.